

Network Security

What is Network Security?

Network Security is the practice of protecting computer networks, devices, applications, and data from unauthorized access, attacks, misuse, modification, or destruction.

Its main objectives are:

- **Confidentiality** – Data should only be accessible to authorized users.
- **Integrity** – Data should not be altered without permission.
- **Availability** – Network services and resources should remain accessible when needed.

These three principles are commonly known as the **CIA Triad**.

Why Network Security is Important

Without proper security, attackers can:

- Steal sensitive information
- Intercept communications
- Modify data
- Spread malware
- Cause service outages
- Gain unauthorized access to systems

Examples:

- Banking transactions
- Online shopping
- Cloud services
- Company networks
- Government systems

Common Network Threats

1. Malware

Malicious software designed to harm systems.

Types:

- Virus
- Worm
- Trojan Horse
- Spyware
- Ransomware

How it works

1. Malware enters through downloads, emails, or infected devices.
2. Executes malicious code.
3. Steals data, damages files, or spreads to other systems.

2. Phishing Attack

Attackers trick users into revealing credentials.

How it works

1. User receives a fake email.
2. Clicks malicious link.
3. Enters username/password.
4. Attacker gains access.

3. Man-in-the-Middle (MITM) Attack

An attacker secretly intercepts communication between two parties.

How it works

Client ----> Attacker ----> Server

Instead of communicating directly:

Client ----> Server

The attacker:

- Reads messages
- Modifies messages
- Steals credentials

4. Denial of Service (DoS)

Attacker floods a server with excessive requests.

How it works

Many Requests



Server



Overloaded

Result:

- Legitimate users cannot access services.

DDoS (Distributed DoS)

Thousands of compromised devices attack simultaneously.

5. Password Attacks

Methods include:

- Brute Force
- Dictionary Attack
- Credential Stuffing

How it works

Attacker repeatedly tries passwords until one succeeds.

Security Mechanisms

1. Authentication

Verifies identity.

Examples:

- Username and Password
- OTP
- Biometrics
- Multi-Factor Authentication (MFA)

Working

User → Login Credentials → Verification → Access Granted

2. Authorization

Determines what authenticated users can access.

Example:

	User	Access
Admin		Full
Employee		Limited
Guest		Minimal

3. Encryption

Converts readable data (plaintext) into unreadable data (ciphertext).

Plaintext

Hello

Ciphertext

X7#P9@

Only authorized parties with the key can decrypt it.

Types of Encryption

A. Symmetric Encryption

Same key for encryption and decryption.

Plaintext

↓

Encrypt(Key)

↓

Ciphertext

↓

Decrypt(Same Key)

↓

Plaintext

Examples:

- AES
- DES

Advantages

- Fast
- Efficient

Disadvantages

- Secure key sharing is difficult

B. Asymmetric Encryption

Uses two keys:

- Public Key
- Private Key

Working

Public Key → Encrypt

Private Key → Decrypt

Examples:

- RSA
- ECC

Advantages

- Secure key exchange

Disadvantages

- Slower than symmetric encryption

Firewall

A Firewall acts as a security barrier between trusted and untrusted networks.

Working

Internet



Firewall



Internal Network

Firewall rules determine:

- Allow traffic
- Block traffic

Examples:

- Block port 23 (Telnet)
- Allow HTTPS (443)

Intrusion Detection System (IDS)

Monitors network traffic for suspicious activities.

Working

Network Traffic



IDS



Alert Generated

IDS only detects and alerts.

Intrusion Prevention System (IPS)

Detects and actively blocks attacks.

Working

Traffic



IPS



Block Malicious Packets

Difference:

IDS

Detects
Alert only

IPS

Detects + Blocks
Prevents attack

Virtual Private Network (VPN)

Creates an encrypted tunnel over a public network.

Working

User



Encrypted Tunnel



VPN Server



Internet

Benefits:

- Privacy
- Secure remote access
- Protection on public Wi-Fi

SSL/TLS Security

Protocols used to secure web communication.

Used in:

<https://>

instead of

<http://>

Working

1. Browser contacts server.
2. Server sends digital certificate.
3. Keys are exchanged securely.
4. Encrypted communication begins.

Browser ⇄ TLS Encryption ⇄ Server

Digital Signatures

Used to verify:

- Sender identity
- Data integrity

Working

1. Sender generates hash.
2. Encrypts hash with private key.
3. Receiver decrypts using public key.
4. Compares hashes.

If hashes match:

- Message is authentic
- Data was not modified

Access Control Lists (ACL)

Rules that control network traffic.

Example:

Allow 192.168.1.10

Deny 192.168.1.20

Routers and firewalls use ACLs to permit or deny packets.

Secure Communication Process (End-to-End Example)

Suppose you log into a banking website:

1. Browser requests website.
2. TLS handshake establishes secure session.
3. Authentication verifies user credentials.
4. Data is encrypted.
5. Firewall filters malicious traffic.
6. IDS/IPS monitors suspicious activity.
7. Secure transaction occurs.

User



TLS Encryption

↓

Firewall

↓

Authentication

↓

Bank Server

Best Practices for Network Security

1. Use strong passwords.
2. Enable MFA.
3. Keep software updated.
4. Use firewalls.
5. Encrypt sensitive data.
6. Regularly monitor logs.
7. Install antivirus software.
8. Restrict unnecessary access.
9. Use VPN for remote access.
10. Backup important data.

Interview Questions

What is the CIA Triad?

Confidentiality, Integrity, and Availability.

Difference between IDS and IPS?

IDS detects attacks and alerts; IPS detects and blocks attacks.

What is a Firewall?

A device/software that filters incoming and outgoing network traffic based on security rules.

Difference between Symmetric and Asymmetric Encryption?

Symmetric uses one key for both encryption and decryption; Asymmetric uses public and private key pairs.

Why is HTTPS secure?

Because it uses SSL/TLS encryption to protect communication between client and server.

What is a VPN?

A technology that creates an encrypted tunnel over a public network for secure communication.

For networking interviews and understanding modern systems, focus especially on **TLS/HTTPS, Firewalls, VPNs, Authentication, Authorization, Encryption, IDS/IPS, and common attacks (MITM, DDoS, Phishing)**. These topics appear frequently in both interviews and real-world network security implementations.